

Chapter

Perspective Chapter: An Overview of Time Series Decomposition and Its Applications

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Abstract

Time series (TS) data is ubiquitous in various fields such as finance, economics, meteorology, and engineering. The analysis of TS data aims to understand the underlying patterns, make predictions, and inform decision-making. One of the fundamental techniques in TS analysis is decomposition, which breaks down a TS into its constituent components: trend, seasonality, and residuals. This chapter provides a comprehensive overview of TS decomposition, breaking down data into trend, seasonality, and residuals. It covers classical methods, such as additive and multiplicative models, advanced techniques like X-12-ARIMA and Seasonal-Trend decomposition using LOESS (STL), and recent approaches, including machine learning (ML) based decompositions. Practical applications in agriculture, meteorology, and economics, along with challenges like non-stationarity and nonlinear behavior, are discussed. The chapter offers guidelines for selecting appropriate methods and includes case studies for real-world insights. It is a valuable resource for researchers, data scientists, and professionals analyzing complex TS data.

Keywords: decomposition methods, residuals, seasonality, statistical analysis, time series analysis, trend analysis

1. Introduction

TS analysis is a fundamental tool in understanding and predicting temporal data, where observations are collected sequentially over time. One of the most critical techniques within this domain is TS decomposition. TS decomposition involves breaking down a complex TS into its underlying components: trend, seasonality, and residuals (or noise). This process provides a clearer understanding of the patterns within the data, enabling more accurate forecasting and analysis [1]. Decomposition is essential for several applications. It enhances the accuracy of predictive models by isolating and understanding each component separately. Additionally, it aids in anomaly detection, where deviations from expected patterns are more easily identified [2]. In various industries, from finance to healthcare, TS decomposition helps identify and interpret trends, optimize resource allocation, and improve decision-making processes. Decomposition techniques help analysts and researchers to gain deeper insights into their data, leading to more informed and effective strategies.

2. What is time series decomposition?

Decomposition is a technique that breaks down complex structures or processes into simpler subcomponents. In the realm of data analysis, it specifically involves separating various factors that influence the behavior of a dataset. TS decomposition is a particular application of this technique, concentrating on TS data. It involves breaking down a TS into several key components to better understand its underlying patterns and behaviors in the data and enhance forecasting accuracy. The primary objective of TS decomposition is to isolate the different factors influencing the data, allowing for more effective modeling and interpretation of the TS.

A TS can be divided into four key components: the trend, seasonal component, cyclic component, and the residual or irregular component.

The trend (T) is a fundamental aspect of TS data, indicating the long-term movement or direction of the data over a specific period. It illustrates whether the data is increasing, decreasing, or remaining stable. Trends can be discerned through graphical representations, line graphs, or statistical techniques like moving averages or regression analysis. There are three trends: upward trend, characterized by an increase over time; downward trend, characterized by a decrease over time; horizontal trend, indicating little to no change, reflecting stability.

The seasonal component (S) highlights regular patterns over fixed intervals (e.g., monthly or quarterly). These patterns manifest consistently due to periodic occurrences such as holidays, climate variations, or business cycles. For instance, retail jacket sales typically rise during the winter season annually, showcasing this cyclical nature of seasonality. Techniques such as seasonal decomposition of TS, seasonal indices, or statistical methods like Fourier analysis and periodograms can help identify seasonal components.

The cyclic component (C) reflects long-term oscillations in the data, often influenced by economic, business, or environmental factors. Unlike seasonality, these cycles vary significantly in duration and do not adhere to fixed periods. They may extend over multiple years and are frequently associated with economic trends. An example would be the fluctuations in business performance during recession or boom periods. Advanced statistical tools, such as regression or spectral analysis, are employed to find these long-term variations and differentiate them from trends and seasonal effects. In many decompositions, the cyclic component is grouped with the trend or considered part of the residuals, depending on the method.

Finally, the residual or noise (R) component represents random fluctuations in the data stemming from unpredictable events, which cannot be accounted for by the trend, seasonality, or cyclic components. These short-term variations may arise from sudden market shifts, natural disasters, or random errors during data collection. The random component is derived by isolating the other components (trend, seasonal, and cyclic) from the data, enabling analysts to pinpoint the remaining random variations.

3. Types of time series decompositions

TS decomposition can be approached using different methods, depending on the data's nature and the analysis's specific objectives. The primary types of TS decomposition are [1, 3].

3.1 Additive decomposition

In additive decomposition, a TS is assumed to be the sum of its components:

$$Y(t) = T(t) + S(t) + R(t) \quad (1)$$

Trend ($T(t)$): indicates the long-term progression or data direction.

Seasonality ($S(t)$): captures regular, repeating patterns within specific intervals, viz. daily, weekly, or annual cycles.

Residual ($R(t)$): accounts for random noise or irregular variations not explained by trend or seasonality.

Additive decomposition is particularly useful for TS, where the seasonal and irregular fluctuations are consistent over time. This method is often applied in scenarios where the data does not exhibit exponential growth or where the impact of seasonal variations is independent of the overall series level.

3.2 Multiplicative decomposition

In multiplicative decomposition, the TS is assumed to be the product of its components:

$$Y(t) = T(t) \times S(t) \times R(t) \quad (2)$$

Trend ($T(t)$): as in additive decomposition, this reflects the underlying direction or pattern.

Seasonality ($S(t)$): here, seasonal variations are proportional to the level of the trend, meaning they increase or decrease in magnitude as the trend changes.

Residual ($R(t)$): represents the random or irregular fluctuations in the data.

Multiplicative decomposition is used when the magnitude of seasonal and irregular variations is related to the trend level. This method is well-suited for TS, where fluctuations increase or decrease proportionally to the trend. For instance, a multiplicative model is more appropriate in economic data such as GDP, where higher GDP levels might coincide with larger seasonal effects.

3.3 Log transformation and decomposition

Sometimes, TS data are neither purely additive nor purely multiplicative. In such cases, a log transformation can be applied to convert a multiplicative model into an additive one:

$$\log(Y(t)) = \log(T(t)) + \log(S(t)) + \log(R(t)) \quad (3)$$

In cases where a TS does not fit neatly into either an additive or multiplicative model, a log transformation can be employed. Applying a logarithm to the data transforms a multiplicative series into an additive one, making it easier to decompose and analyze. This approach is instrumental in financial data, such as stock prices or sales figures, where the data might exhibit additive and multiplicative effects. After decomposition on the log-transformed data, the components can be exponentiated to revert to the original scale.

3.4 STL decomposition (seasonal and trend decomposition using loess)

STL is a robust and versatile method of decomposing TS into trend, seasonality, and residual components. It is beneficial for handling complex TS with multiple seasonal cycles or irregular patterns.

Seasonal: extracted using LOESS (Locally Estimated Scatterplot Smoothing), which can adapt to changing seasonal patterns.

Trend: also smoothed using LOESS, providing a flexible approach to capturing the long-term movement.

Residual: the remaining component after removing the seasonal and trend effects.

STL decomposition offers a flexible and robust way to break down TS data, particularly when dealing with complex or irregular seasonal patterns. Unlike classical methods, STL does not assume that seasonality and trend are constant over time, making it adaptable to changes in the underlying data structure. STL is particularly effective for analyzing TS with multiple seasonal components, such as daily and yearly cycles in climate data, or for data with nonlinear trends. Customizing the smoothing parameters allows analysts to capture subtle variations in the data, making STL decomposition a powerful tool for both exploratory data analysis and forecasting.

3.5 Classical decomposition

Classical decomposition methods are based on simple moving averages to estimate trends and seasonality.

Trend: estimated by smoothing the series using a moving average filter, removing short-term fluctuations.

Seasonality: determined by subtracting the trend from the original series and isolating the seasonal pattern.

Residual: represents the difference between the original series and the combined trend and seasonality components.

Classical decomposition methods are based on moving averages and are among the earliest techniques developed for TS analysis. While simple, they are effective for TS with clear and stable seasonal patterns. In classical decomposition, the trend is typically extracted using a centered moving average, which smooths out short-term fluctuations.

The seasonal pattern is identified by calculating the average of the detrended data for each season. The remaining part, the difference between the original data and the combined trend and seasonal components, is considered the residual. This method is useful for its simplicity and ease of interpretation, making it a good starting point for TS analysis.

3.6 X-11 and X-13-ARIMA-SEATS decomposition

These are advanced statistical techniques used mainly for economic and financial TS. They provide detailed seasonal adjustments and are widely used by government agencies for official statistics.

X-11: an earlier method that adjusts for seasonal effects and trading days.

X-13-ARIMA-SEATS: a more advanced approach that integrates ARIMA modeling with the decomposition process to improve forecasting accuracy.

These methods are particularly valuable when dealing with the data's complex seasonal patterns and irregularities. These methods are particularly valuable in official statistics, where accurate seasonal adjustment is crucial for making informed policy decisions and economic analyses.

4. Importance of time series decomposition

TS decomposition is a critical technique in TS analysis that plays a fundamental role in understanding, interpreting, and forecasting temporal data. Here are the key reasons why TS decomposition is important [1:3].

4.1 Enhanced understanding of data

TS decomposition breaks down a complex TS into its basic components—trend, seasonality, and residuals. This separation provides a clearer picture of the underlying hidden patterns in the data. By differentiating these individual components, analysts can better understand the long-term trend, cyclical patterns, and random variations within the series. This understanding is essential for making informed decisions based on the data.

4.2 Improved forecasting accuracy

One of the primary applications of TS decomposition is forecasting. A breakdown of the TS allows for the analysis and forecasting of each component separately. Combining these predictions allows forecasting models to generate more precise and trustworthy forecasts. For instance, understanding seasonal patterns allows for more precise predictions during peak periods, while trend analysis can provide insight into long-term growth or decline.

4.3 Facilitating seasonal adjustment

Eliminating seasonal effects from TS data reveals the underlying trend in many fields, especially in economics and finance. This process is known as seasonal adjustment. It plays an important role in accurate analysis and decision-making. For example, unemployment rates often vary seasonally, and decomposing the series helps analysts remove these fluctuations to focus on the true trend.

4.4 Detection of anomalies or outliers

Decomposing a TS makes detecting anomalies or outliers in the data easier. Analysts can identify unusual deviations from expected patterns by isolating the residual component, which represents the noise or unexplained variation. This is particularly valuable in fields like finance, where detecting anomalies can signal fraud or errors, or in manufacturing, where it can indicate quality control issues.

4.5 Better model selection and validation

Understanding a TS's components helps select the appropriate models for analysis and forecasting. For instance, knowing whether a series has a strong seasonal

component might suggest using models like seasonal ARIMA (SARIMA) or exponential smoothing methods. Additionally, decomposition allows for better validation of models by comparing how well they capture each component, leading to more robust and reliable models.

4.6 Simplifying complex data

Real-world TS data can be complex, with multiple overlapping patterns. Decomposition simplifies this complexity by breaking the series into manageable components. This simplification not only aids in analysis but also in communicating insights to stakeholders who may not be familiar with advanced statistical methods. Analysts can convey key insights more effectively by presenting the data in terms of trend, seasonality, and residuals.

4.7 Guiding business strategy

Understanding the components of a TS can inform strategic decisions for businesses. For example, identifying a strong upward trend might lead to increased investment in capacity, while understanding seasonal patterns can guide marketing and inventory decisions. By decomposing TS data, businesses can align their strategies with the underlying dynamics of their market.

4.8 Noise reduction

Decomposition helps in filtering out noise from the TS data. The residual component, which represents the noise, can be isolated and analyzed separately. This noise reduction is particularly important in sensitive applications, such as control systems or real-time monitoring, where noise can obscure important signals.

4.9 Supporting multivariate analysis

In cases where multiple TS are being analyzed simultaneously, decomposition can compare trends and seasonal patterns across different series. This is particularly useful in economic analysis, where different indicators may need to be compared, or in marketing, where sales data across different regions or products might be analyzed in parallel.

5. Recent developments in time series decompositions

TS decomposition has evolved significantly in recent years with the advancements in computational methods, the availability of big data, and the growing complexity of real-world applications. These developments have introduced new techniques and improved existing methods, making TS decomposition more powerful, flexible, and applicable to a broader range of problems [4]. Here are some of the key recent developments in TS decomposition.

5.1 Advanced machine learning techniques

Recent innovations in ML have led to the development of novel decomposition methods that leverage deep learning architectures and advanced algorithms.

For instance, techniques like long short-term memory (LSTM) networks and Convolutional Neural Networks (CNNs) have been adapted to TS decomposition. These can read complex, nonlinear patterns and interactions in TS data that traditional methods might miss. For an example:

Seasonal-Trend decomposition using LSTM (STL-LSTM): This approach combines the power of LSTM networks with traditional STL decomposition, allowing for the modeling of nonlinear trends and seasonality. The LSTM component captures long-term dependencies and nonlinearities, while the STL part handles the decomposition of the TS into TS components, viz. trend, seasonal, and residual.

5.2 Bayesian time series decomposition

Bayesian methods have been increasingly applied to TS decomposition. These methods allow for the incorporation of prior knowledge and estimating uncertainty in the decomposition process. For example, Bayesian Structural Time Series (BSTS): This framework uses Bayesian inference to decompose TS into components such as trend, seasonality, and regression effects while also accounting for the uncertainty in each component. BSTS is particularly useful for scenarios where data is sparse or noisy, providing more robust and interpretable decompositions.

5.3 State-space models and Kalman filtering

State-space models have seen renewed interest in TS decomposition, particularly with the integration of Kalman filters. These models provide a dynamic and flexible way to model TS data, allowing for real-time decomposition as new data becomes available. For an example:

Dynamic linear models (DLMs): A specific type of state-space model, DLMs allow for the dynamic updating of trend and seasonal components over time. Kalman filtering estimates these components in real time, making this approach particularly useful for applications like financial forecasting, where data arrives continuously.

5.4 Empirical mode decomposition (EMD) and variants

Empirical mode decomposition (EMD) has been increasingly used in TS analysis, especially in applications where the data is nonlinear and nonstationary. EMD decomposes a TS into intrinsic mode functions (IMFs) without assuming a specific model structure. For an example:

Ensemble Empirical Mode Decomposition (EEMD): EEMD, an improved version of EMD, solves the issue of mode mixing by adding random noise to the data before breaking it down into its components. This approach is particularly useful in biomedical signal processing and climate science, where data can be highly irregular and complex.

5.5 Robust decomposition methods

Robust decomposition methods have gained popularity due to the growing need to handle outliers and noise in TS data. These methods are designed to be less sensitive to anomalies, ensuring that the decomposition process remains stable and reliable despite irregularities. Example:

RobustSTL: An extension of the classical STL (seasonal-trend decomposition using LOESS), RobustSTL is designed to handle outliers and noise more effectively.

It achieves this by incorporating robust statistical techniques into the decomposition process, making it suitable for TS with significant noise or irregular fluctuations.

5.6 Frequency domain decomposition

Recent developments have also focused on frequency domain approaches to TS decomposition, which involve analyzing the spectral properties of the data. These methods are particularly useful for identifying underlying periodic patterns and separating components in the frequency domain. Example:

Wavelet decomposition: This method decomposes a TS into components at different scales, allowing for the analysis of both high-frequency (short-term) and low-frequency (long-term) patterns. Wavelet decomposition is widely used in fields like finance and engineering, where understanding multi-scale phenomena is crucial.

5.7 Tensor decomposition for multivariate time series

Tensor decomposition methods have emerged as powerful tools for scenarios where multiple TS are analyzed simultaneously. Tensor decomposition originates from multilinear algebra (a generalization of matrix decomposition). Tensors, or multidimensional arrays, extend matrices to higher dimensions and can capture complex relationships across multiple factors or modes (e.g., time, space, and variables). Tensor decomposition is a mostly used technique when we collect time series data across multiple variables or dimensions (e.g., time, location, and variable type). Tensor decomposition helps find out important components from high-dimensional data. Tensor decomposition is suitable for capturing patterns, identifying underlying factors, or reducing dimensionality.

These methods extend traditional decomposition techniques to handle multidimensional data, capturing complex relationships across multiple TS. An example is canonical polyadic (CP) decomposition: applied to TS data organized in a tensor format, CP decomposition separates the data into components that capture trends, seasonality, and interactions across different dimensions. This approach is increasingly used in recommendation systems and social network analysis, where multivariate TS data is common. Tensor decompositions can enhance model interpretability, make high-dimensional data more manageable, and improve predictive performance by focusing on the most significant patterns in the data. Tensor decomposition in multivariate time series is widely used in fields such as finance, neuroscience, and sensor networks, where understanding time-dependent relationships across multiple variables is critical.

5.8 Seasonal hybrid extreme decomposition (SHED)

Seasonal hybrid extreme decomposition (SHED) is a technique that analyzes complex seasonal patterns and trends in time series data. SHED is a recent approach combining decomposition techniques to handle complex TS data. It is also known as the ensemble hybrid decomposition technique. In short, SHED combines traditional decomposition methods with advanced hybrid approaches to effectively separate a time series into different components: seasonal, trend, and extreme events. The primary goal of SHED is to enhance forecasting accuracy, especially for data that exhibits seasonal variations and sudden, extreme fluctuations (e.g., spikes or dips). The critical components of SHED are seasonal (daily, weekly, or yearly patterns), trend

(the long-term progression or general direction of the data), and extreme components (sharp peaks, anomalies, or outliers that do not fit into regular seasonal or trend patterns). The approach integrates wavelet transform, EMD, and other methods to separate a TS into more granular components, enhancing the accuracy of subsequent analysis and forecasting. SHED is applicable in fields requiring both regular pattern recognition and robustness to anomalies, ultimately enhancing forecasting and decision-making in dynamic environments like energy demand forecasting, financial market analysis, etc.

6. Application of time series decompositions in agriculture

In agriculture, it has several applications. Agricultural commodity prices are highly volatile, mostly nonlinear and nonstationary, due to several factors like uncertainty in production, demand-supply imbalances, government policy interventions, and others. These make the forecasting of agricultural commodity prices more challenging to the investigators. Among the different methods available in the literature, decomposition-based methods are more important and popular due to their “divide and conquer” principle. The evolutions and uses of decomposition methods mainly depend on several data properties, like seasonal patterns, heterogeneity, linearity, and the stationarity of data.

The STL, a versatile and robust decomposition-based method, is popularly used for agricultural price forecasting [4–7] and for forecasting the commodity prices. First, deseasonalized the series by extracting the seasonal component calculated by STL decomposition. Second, point forecasting of price values is done by constructing statistical models, like ARIMA, ETS, and others, on a seasonally adjusted series. Finally, the integration of seasonal components to get the final forecast value [6]. However, the main challenges for forecasting through simple STL decomposition are mainly due to assumptions of linear trend, additive seasonality, choice of the seasonal cycle length and smoothing parameters, and proper choice of model for forecasting different components.

Several hybrid methods combining STL with ML techniques have been developed and used for forecasting to deal with such problems. For instance, recent developments in STL with ELM (extreme learning machines) are used for seasonal agricultural commodities, particularly for vegetable market price forecasting [5]. Complete detail can be found in a study by Jaiswal et al. [7], where the seasonal TS has been first decomposed into a trend, seasonal, and remainder components. Secondly, each component has been individually forecasted using ELM. Finally, the forecasts of individual components (trend, seasonal, and reminder) were added to produce the final estimates for the original series. A schematic diagram is presented in **Figure 1**. The outputs of the empirical results are best compared to traditional methods for forecasting.

High-frequency TS data often exhibit nonlinear and nonstationary behavior with fluctuations. Methods based on EMD and its variants (hybrid models) are widely used to address these characteristics in agricultural data [8–12]. Analytical R packages for EMD variants, such as EMD-ANN and EMD-SVR, are available in the literature. In a study by Das et al. [10], they applied EMD-ANN and EMD-SVR to forecast the monthly price index of vegetables. Initially, the EMD technique decomposes the original nonlinear and nonstationary data into a limited number of independent sub-series, including k intrinsic mode functions (IMFs) and a residual. These IMFs and the residual are then

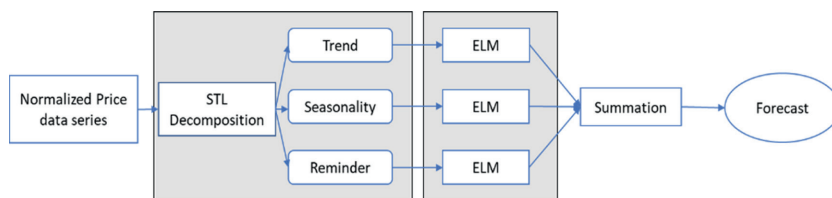


Figure 1.
STL-ELM method for agricultural commodity price forecasting.

modeled and predicted using ANN or SVR. The forecasted values of the IMFs and the residual are summed to produce an ensemble forecast for the original series. When comparing the performance of EMD-based hybrid models with standard models like ANN and SVR, the results indicated that while ANN and SVR are commonly used for nonlinear data, they struggled with the nonstationary nature of the dataset. In contrast, the hybrid models EMD-ANN and EMD-SVR performed better due to their ability to capture both nonlinear and nonstationary patterns, making them a viable alternative for forecasting volatile agricultural price series (**Figure 2**).

Recently, variational modal decomposition (VMD) and its variants have been widely used to address the challenges of highly volatile, nonlinear, and nonstationary agricultural commodity price data [13]. Zhang et al. [14] applied VMD variants to forecast volatile agricultural commodities such as lean hog prices, soybean meal prices, and soybean oil prices in China. They achieved promising results. **Figure 3** provides a schematic representation of this hybrid forecasting approach. This study compares a hybrid model's effectiveness (VMD-ARIMA-LSTM) to seven other forecasting models. These include single models (RF, LSTM, BP), models using EMD decomposition (EMD-ARIMA-SVM, EMD-BP), and models using VMD decomposition (VMD-LSTM, VMD-ARIMA-BP). The results showed that VMD-based models, particularly VMD-ARIMA-LSTM, significantly outperform EMD-based models due

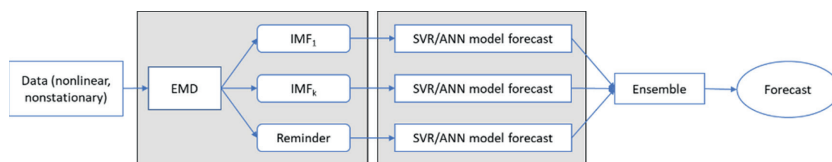


Figure 2.
Hybrid model EMD-SVR/EMD-ANN for agricultural price forecasting.

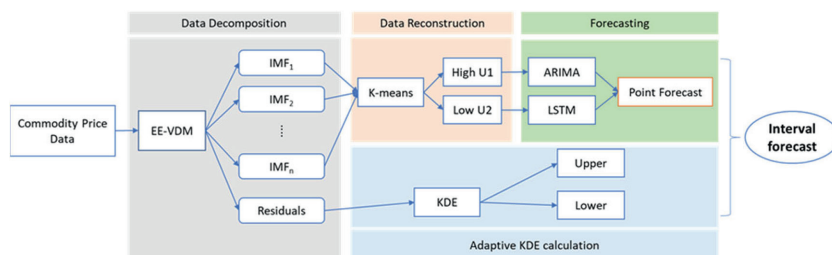


Figure 3.
Hybrid model VMD-ARIMA-LSTM for lean hog prices, soybean meal prices, and soybean oil prices in China.

to VMD's ability to resolve modal confusion and end effects, enhancing decomposition performance. In forecasting experiments on three datasets, VMD-ARIMA-LSTM showed improvements in RMSE by 5.46%, 14.68%, and 10.28% for 1-step, 3-step, and 6-step ahead predictions, respectively, and similarly improved MAPE results. It has been concluded that decomposing and reconstructing the original sequence and selecting an appropriate forecasting model based on sequence complexity leads to better predictions for highly volatile agricultural commodities. This highlights the advantages of the decomposition-integration strategy. A Diebold-Mariano (DM) test further confirmed the superiority of VMD-ARIMA-LSTM. However, due to its lower complexity, no significant improvement has been observed for lean hog price data in certain steps. Overall, the decomposition-based models are practical and applicable to agricultural price forecasting.

7. Applications of time series decompositions in other sectors

The TS decomposition methods are widely applied in various fields beyond agriculture, like Environmental Science, Meteorology, and Climate Science [3, 15–18], Economics and Finance [1, 19], Marketing and Sales, Engineering, and Manufacturing [20–22], Healthcare and Epidemiology [23, 24], and many other fields, to understand forecast, and control time-dependent data.

The uses of TS decomposition in environmental science are to forecast key variables like temperature, rainfall, pollution levels, or other meteorological parameters. These parameters are the key indicators of climate change, and the TS decompositions are used for accurate forecasting. In recent times, hybrid methods have been more popularly used, rather than the traditional TS decomposition methods, like the DL-MEMD model for hourly global horizontal irradiance forecasting [25], Hybrid DL-MVMD model for estimating daily solar radiation [26], Hybrid gray-STL with Gaussian process for CO₂ emissions forecasting [18]. Decomposition techniques like STL and its variants are commonly used to forecast temporal features of meteorological data. A study by Liu and Zhang [27] used a decomposition-based prediction model combining STL and gated recurrent units (STL-GRU) for forecasting the surface temperature of various regions in China. This method analyzed these regions' patterns and long-term trends in temperature data.

The decomposition techniques are also popularly used for precipitation forecasting, especially when the chances of erroneously mixing future information into the training data occur. Jiao and He [28] effectively employed the VMD-BiLSTM-based stepwise decomposed ensemble method to overcome the mixing problem for practical forecasting of reciprocation. A study by Parviz and Ghorbanpour [29] improved hybrid models by integrating linear and nonlinear approaches focusing on signal decomposition for complex nonlinear components. It evaluates the effectiveness of combining SARIMA with EMD and maximal overlap discrete wavelet transform (MODWT) utilizing monthly rainfall data from Iran. The process involved using SARIMA to model the data, decomposing its error series into IMFs with EMD, and then forecasting these IMFs using support vector regression.

TS decomposition is widely used to forecast economic indicators such as inflation, GDP, and unemployment rates. Analysts can make more precise predictions about future economic conditions by separating the trend (long-term growth), seasonal effects (like holiday impacts), and random variations. For instance, tools like ARIMA and seasonal decomposition of time series (STL) filter out the seasonal

component and capture core economic trends. In stock market forecasting, decomposition methods help identify the underlying price trend, seasonal behavior (such as quarterly earnings reports), and residual noise. For example, EMD has been used in currency exchange forecasting to identify local trends and reduce the noise before applying predictive models like least squares support vector regression [30]. A study by Sohrabbeig et al. [31] extended beyond single-seasonal patterns, incorporating methods like Multiple Seasonal-Trend Decomposition using Loess (MSTL). MSTL can handle complex datasets with multiple seasonal cycles, such as daily and weekly patterns in financial TS. This is particularly useful for high-frequency trading, where market behavior follows distinct patterns over different time horizons. The multiresolution analysis through wavelets is increasingly popular for nonstationary TS, as is stock prices and volatility [30, 32]. By decomposing the series into different frequency components, this method helps capture both short-term fluctuations and long-term trends, enabling robust modeling of dynamic financial systems.

TS decomposition is a versatile technique with applications across multiple fields. Breaking down complex data into components provides insights into long-term trends, cyclical variations, and irregularities, supporting better decision-making and planning in various industries.

8. Packages/software of TS decomposition

R/RStudio, Python, and MATLAB are software that is primarily used for TS decomposition. There are packages in the software for applying a specific TS decomposition technique (Table 1).

9. Conclusions

In conclusion, TS decomposition is a critical technique for understanding and analyzing complex data across various fields, including finance, economics, meteorology, and engineering. This chapter has outlined both classical and advanced decomposition methods, such as additive, multiplicative, STL, and newer ML approaches,

Method	R package
VMD	VMDDecomp
EMD	EMD
EEMD	EMD
Wavelet	wavelets
Fourier decomposition	swdft
Tensor decomposition	rTensor
Seasonal and Trend decomposition	stl
Kalman filter	Rcpp
Bayesian time series decomposition	Rbeast

Table 1.
Some decomposition techniques and required R packages for their application.

highlighting their practical applications in diverse domains like agriculture and economics. Additionally, the chapter addresses key challenges, such as non-stationarity and nonlinearities, offering a thorough guide for selecting appropriate methods and understanding real-world TS data through case studies. In the end, this chapter provides a complete overview of TS decomposition and its practical applications for forecasting to inform data-driven decision-making.

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Conflict of interest

The authors declare no conflict of interest.

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References

- [1] Box GE, Jenkins GM, Reinsel GC, Ljung GM. Time Series Analysis: Forecasting and Control. 5th ed. New Jersey: Wiley; 2015
- [2] Shumway RH, Stoffer DS. Time Series Analysis and Its Applications With R Examples. 4th ed. Switzerland: Springer; 2017
- [3] Chatfield C. The Analysis of Time Series: Theory and Practice. London: Chapman and Hall; 2013
- [4] Cleveland RB, Cleveland WS, McRae JE, Terpenning IJ. STL: A seasonal-trend decomposition procedure based on loess. *Journal of Official Statistics*. 1990;1:3-33
- [5] Xiong T, Li C, Bao Y. Seasonal forecasting of agricultural commodity price using a hybrid STL and ELM method: Evidence from the vegetable market in China. *Neurocomputing*. 2018;275:2831-2844
- [6] Ouyang Z, Ravier P, Jabloun M. STL decomposition of time series can benefit forecasting done by statistical methods but not by machine learning ones. *Engineering Proceedings*. 2021;5(1):42
- [7] Jaiswal R, Choudhary K, Kumar RR. STL-ELM: A decomposition-based hybrid model for price forecasting of agricultural commodities. *National Academy Science Letters*. 2022;45(6):477-480
- [8] Das P. Study on machine learning techniques based hybrid model for forecasting in agriculture [thesis]. New Delhi: Graduate School, Indian Agricultural Research Institute; 2019
- [9] Das P, Jha GK, Lama A, Parsad R, Mishra D. Empirical mode decomposition based support vector regression for agricultural Price forecasting. *Indian Journal of Extension Education*. 2020;56(2):7-12
- [10] Das P, Jha GK, Lama A. Empirical mode decomposition based hybrid machine learning models for agricultural commodity price forecasting. *Statistics and Applications*. 2023;21(1):99-112
- [11] Choudhary K, Jha GK, Jaiswal R, Kumar RR. Decomposition-based long short-term memory model for price forecasting of agricultural commodities. *Iran Journal of Computer Science*. 2024;2:1-3
- [12] Ghose B, Pandit P, Mazumder C, Sinha K, Sahu PK. Comparative study of EMD based modelling techniques for improved agricultural price forecasting. *Journal of the Indian Society of Agricultural Statistics*;78(1):53-62
- [13] Das P, Jha GK, Lama A. Variational mode decomposition based machine learning models optimized with genetic algorithm for price forecasting. *Journal of the Indian Society of Agricultural Statistics*. 2022;76(3):141-150
- [14] Zhang D, Zhang X, Hu H, Zhang B, Ling L. A hybrid model for point and interval forecasting of agricultural price based on the decomposition-ensemble and KDE. *Soft Computing*. 2024;28(17):10153-10176
- [15] Chatfield C, Yar M. Holt-winters forecasting: Some practical issues. *Journal of the Royal Statistical Society Series D: The Statistician*. 1988;37(2):129-140
- [16] Zhang X, Zwiers FW, Li G. Monte Carlo experiments on the detection of trends in extreme values. *Journal of Climate*. 2004;17(10):1945-1952

- [17] Moreno-Carbonell S, Sanchez-Ubeda EF, Muñoz A. Time series decomposition of the daily outdoor air temperature in Europe for long-term energy forecasting in the context of climate change. *Energies*. 2020;**13**(7):1569
- [18] Yuan H, Ma X, Ma M, Ma J. Hybrid framework combining grey system model with Gaussian process and STL for CO2 emissions forecasting in developed countries. *Applied Energy*. 2024;**360**:122824
- [19] Souropanis I, Vivian A. Forecasting realized volatility with wavelet decomposition. *Journal of Empirical Finance*. 2023;**74**:101432
- [20] Montgomery DC. Introduction to Statistical Quality Control. United Kingdom: John Wiley & Sons; 2013
- [21] Hyndman RJ. Forecasting: Principles and Practice. USA: OTexts; 2018
- [22] Qiu Y, He Z, Zhang W, Yin X, Ni C. MSGCN-ISTL: A multi-scaled self-attention-enhanced graph convolutional network with improved STL decomposition for probabilistic load forecasting. *Expert Systems with Applications*. 2024;**238**:121737
- [23] Nsoesie EO, Brownstein JS, Ramakrishnan N, Marathe MV. A systematic review of studies on forecasting the dynamics of influenza outbreaks. *Influenza and Other Respiratory Viruses*. 2014;**8**(3):309-316
- [24] Baharom M, Ahmad N, Hod R, Abdul Manaf MR. Dengue early warning system as outbreak prediction tool: A systematic review. *Risk Management and Healthcare Policy*. 2022;**15**:871-886
- [25] Gupta P, Singh R. Combining a deep learning model with multivariate empirical mode decomposition for hourly global horizontal irradiance forecasting. *Renewable Energy*. 2023;**206**:908-927
- [26] Band SS, Qasem SN, Ameri R, Pai HT, Gupta BB, Mehdizadeh S, et al. Deep learning hybrid models with multivariate variational mode decomposition for estimating daily solar radiation. *Alexandria Engineering Journal*. 2024;**105**:613-625
- [27] Liu X, Zhang Q. Combining seasonal and trend decomposition using LOESS with a gated recurrent unit for climate time series forecasting. Vol. 12. *IEEE Access*; 2024. pp. 85275-85290
- [28] Jiao X, He Z. A novel coupled rainfall prediction model based on stepwise decomposition technique. *Scientific Reports*. 2024;**14**(1):10853
- [29] Parviz L, Ghorbanpour M. A hybrid EMD and MODWT models for monthly precipitation forecasting using an innovative error decomposition method. *Stochastic Environmental Research and Risk Assessment*. 2024;**14**:1-24
- [30] Rabbouch H, Rabbouch B, Saâdaoui F. Multiresolution data analytics for financial time series using MATLAB. In: *Data Analytics for Management, Banking and Finance: Theories and Application*. Switzerland: Springer Nature; 2023. pp. 113-134
- [31] Sohrabbeig A, Ardakanian O, Musilek P. Decompose and conquer: Time series forecasting with multiseasonal trend decomposition using loess. *Forecasting*. 2023;**5**(4):684-696
- [32] Berger T. Forecasting based on decomposed financial return series: A wavelet analysis. *Journal of Forecasting*. 2016;**35**(5):419-433